

## Producer

```
for (int n = 0; n < 1; n += 1)
  for (int o = 0; o < 64; o += 1)
    for (int h = 0; h < 56; h += 1)
      for (int w = 0; w < 56; w += 1)
        for (int kh = 0; kh < 3; kh+=1)
          for (int kw = 0; kw < 3; kw+=1)
            out[n][o][h][w] = max(...);
```

## Consumer

```
for (int n = 0; n < 1; n += 1)
  for (int o = 0; o < 64; o += 1)
    for (int h = 0; h < 56; h += 1)
      for (int w = 0; w < 56; w += 1)
        ... = out[n][o][h][w];
```



Access Count: 1x64x56x56x3x3 **✗** Access Count: 1x64x56x56

Index dim: **n, o, h, w**

Reduction dim: **kh, kw**

## Producer Rewritten

```
for (int n = 0; n < 1; n += 1) {
  for (int o = 0; o < 64; o += 1)
    for (int h = 0; h < 56; h += 1)
      for (int w = 0; w < 56; w += 1){
        int temp = 0;
        for (int kh = 0; kh < 3; kh+=1){
          for (int kw = 0; kw < 3; kw+=1)
            temp = max(...);
        }
        out[n][o][h][w] = temp;
      }
}
```



Access Count: 1x64x56x56 **✓** Access Count: 1x64x56x56